

Exponential Distribution

What does exponential distribution model?

- Exponential distribution is used to model the time until one event occurs.
- Exponential distribution is used to model the time between two independent events.
- Note that number of independent event that occur in a fixed amount of time has Poisson distribution, but the time between two independent event has exponential distribution.
- Exponential distribution also models the life time of electronics devices, for example life time of a bulb.

Definition (Exponential distribution $X \sim \text{Exp}(\lambda)$)

X has an exponential distribution with parameter $\lambda > 0$ if

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$E[X] = \frac{1}{\lambda}$$

$$\text{Var}[X] = \frac{1}{\lambda^2}$$

Note. λ is the average rate of occurrence per unit time. That is same occurrence rate we were using in Poisson distribution.

Eg 1. Suppose that you create a YouTube channel and your channel gets an average of 4 unique visits per hour. What is the probability that your channel will go more than 20 minutes without a visitor?

method one:

$$\lambda = 4 \quad \frac{\text{visit}}{\text{hour}}$$

$$20 \text{ minutes} = \frac{1}{3} \text{ hour}$$

$$T \sim \text{Exp}(4)$$

$$P(T > \frac{1}{3}) = \int_{\frac{1}{3}}^{\infty} \lambda e^{-\lambda t} dt = -e^{-\lambda t} \Big|_{\frac{1}{3}}^{\infty}$$

$$= -\lim_{t \rightarrow +\infty} e^{-\lambda t} + e^{-\frac{\lambda}{3}} = e^{-\frac{\lambda}{3}}$$

0

Eg 1. Suppose that you create a YouTube channel and your channel gets an average of 4 unique visits per hour. What is the probability that your channel will go more than 20 minutes without a visitor?

method 2: As we did in the lecture

$$\lambda = 4 \quad \frac{\text{visit}}{\text{hour}} \quad \xrightarrow{\text{for minute}}$$

$$\lambda_1 = \frac{4}{60} \quad \frac{\text{visit}}{\text{minute}}$$
$$\lambda_1 = \frac{1}{15}$$

$$T \sim \text{EXP}\left(\frac{1}{15}\right)$$

$$\begin{aligned} P(T > 20) &= \int_{20}^{\infty} \lambda_1 e^{-\lambda_1 t} dt = -e^{-\lambda_1 t} \Big|_{20}^{\infty} \\ &= 0 + e^{-20 \lambda_1} \\ &= e^{-\frac{20}{15}} = e^{-\frac{4}{3}} \end{aligned}$$

Definition (CDF of Exponential Distribution)

Suppose $X \sim \text{Exp}(\lambda)$, then its CDF $F(x)$ is given by

$$F(x) = P(X \leq x) = \begin{cases} 1 - e^{-\lambda x} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

Derivation:

$$\text{If } a \leq 0 \quad F(a) = \int_{-\infty}^a f(x) dx = 0$$

$$\text{If } a > 0 \quad F(a) = \int_{-\infty}^a f(x) dx = \int_0^a \lambda e^{-\lambda x} dx$$

$$F(a) = -e^{-\lambda x} \Big|_0^a = e^0 - e^{-\lambda a} = 1 - e^{-\lambda a}$$

$$\Rightarrow F(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ 1 - e^{-\lambda x} & \text{if } x > 0 \end{cases}$$

Eg 2. Find the 3rd quartile of exponential distribution. **Recall** the point x is the third quartile if $F(x) = \frac{3}{4}$.

$$\text{3rd Quartile} = a \Rightarrow P(X \leq a) = \frac{3}{4} \Rightarrow F(a) = \frac{3}{4}$$

$$X \sim \text{Exp}(\lambda) \xRightarrow{\text{by definition}} F(a) = \frac{3}{4} = 1 - e^{-\lambda a}$$

$$e^{-\lambda a} = \frac{1}{4} \Rightarrow -\lambda a = \ln\left(\frac{1}{4}\right) \Rightarrow$$

$$-\lambda a = \ln(1) - \ln 4 \Rightarrow a = \frac{\ln 4}{\lambda}$$